

Generative AI

What is Generative AI?

- Generative AI: Creates new, original content.
 - Examples: ChatGPT, Gemini, Grok, DeepSeek.
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- Goal: Understand practical applications, thrive in the AI age.

Your Genius Colleague

- Access to sum of human knowledge.
- Limitations: Can make mistakes, misunderstand.
- Crucial Skill: Prompt Engineering / Prompt Design.
- Vastly underestimated capabilities.

AI vs. Generative AI

- Artificial Intelligence (AI): Broad field, decades old.
- Examples: YouTube recommendations, fraud detection, search results.
- Generative AI: Specific type of AI that generates new, original content.
- Generates new, original content. (The "G" in GPT).

Large Language Models (LLMs)

- Type of Generative AI.
- Communicate using normal human language.
- Based on Transformer architecture (The "T" in GPT).

LLMs: The "Guess the Next Word" Machine

- Core: Artificial Neural Network (billions/trillions of parameters).
- Content (text, images, audio) represented as numbers.
- Input text -> Numbers -> Neural Network -> Output Numbers -> Output Text
- Example: "Dogs are..." -> "animals."

Training the Model: Pre-training

- Fed massive amounts of text data (internet, books, articles)
- Plays "guess the next word" repeatedly.
- Parameters tweaked via backpropagation until accurate.
- "P" in GPT: Pre-trained.

Training the Model: Human Feedback (RLHF)

- Reinforcement Learning with Human Feedback (RLHF).
- Humans test, evaluate, and provide feedback.
- Aligns model behavior with human values and safety.
- Example: Model won't help commit crimes.

Beyond ChatGPT

- Vary in: Speed, Capability, Cost, Availability, Licensing, Ease of Use, Specialization, Integration.
- Generally, you get what you pay for (e.g., GPT-3.5 vs. GPT-4).

Types of Generative AI Models by Content Type

- Text-to-Text
- Text-to-Image
- Image-to-Image
- Image-to-Text
- Speech-to-Text
- Text-to-Audio
- Text-to-Video

Text-to-Text Models

- Input: Text, Output: Text (natural language, code, JSON, HTML).
- Use Cases:
 - Code Generation (snippets, debugging, translation).
 - Content Creation (emails, summaries, translation).
 - Educational Use (quiz questions, lesson plans, rubrics, feedback).
 - Customer Service (chatbots).

Text-to-Image Models

- Input: Text description, Output: Image.
- Examples: DALL-E, Midjourney, Stable Diffusion.
- Use Cases:
 - Concept Art (games, film).
 - Marketing & Advertising (unique visuals).
 - Personalized Content (illustrations, greeting cards).
 - Education (visual aids, historical events, scientific concepts).

Image-to-Image Models

- Input: Image, Output: Transformed/new image.
- Use Cases:
 - Image Editing & Enhancement (background removal, upscaling, variations).
 - Design Iteration (multiple design variations from sketch).
 - Image Suggestion: Two images with an arrow between them, one original, one transformed (e.g., a photo and the same photo in a different art style).

Image-to-Text Models

- Input: Image, Output: Text description of contents.
- Use Cases:
 - Accessibility (alt-text for visually impaired).
 - Content Moderation (identifying inappropriate content).
 - Visual Search (searching images by description).
 - Image Suggestion: An image with a text box next to it containing a description of the image.

Speech-to-Text Models

- Input: Audio, Output: Voice transcription.
- Use Cases:
 - Meeting Notes (transcribing conversations).
 - Podcast & Video Transcription (searchable content).
 - Voice Commands (Siri, Alexa).

Text-to-Audio Models

- Input: Text prompt, Output: Music or sounds.
- Use Cases:
 - Voiceovers (synthetic voices for narrations, audiobooks)
 - Sound Effects (specific sounds for games, videos).
 - Music Composition (short pieces in specified style).

Text-to-Video Models

- Input: Text prompt, Output: Video.
- Use Cases:
 - Short Ads (quick animated advertisements).
 - Animated Explainers (simple concept videos).
 - Personalized Content (future of entertainment?).

Multimodal AI: Combining Capabilities

- Combines different models (text, image, audio, video).
- Example: ChatGPT app (photo input, text output).

The Crossing Point: Human vs. AI Intelligence

- Computer capabilities: Rapidly improving, especially with Generative AI.
- AI now fluent in human language, performing intellectual/creative tasks.
- We are at a crossing point: AI excels at some things, humans at others.
- AI's exponential improvement vs. human stability.

Building with AI: APIs

- Developers use models to build custom AI-powered products/features.
- Your product interacts with AI model via APIs (Application Programming Interfaces).
- AI models act as an "external brain" for your product.

AI, a Threat?

- Generative AI is a super useful tool: Opportunity, not threat.
- More powerful than you think: Limitation is your imagination & prompt skills.
- Prompt Engineering is crucial: Practice to improve.
- Experiment! Make AI part of your daily life.